



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	CHEMISTRY IN DAILY LIFE				
Type of Course	MDC				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	Role of chemicals in or life. Basic idea of environmental pollution.				
Course Summary	This course ensures that the students acquire a profound knowledge and understanding on chemicals that are used in daily life.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	<i>Know the different chemicals that sustain our life</i>	U	C	Instructor-created exams / Quiz
CO2	<i>Understand the role of chemistry in forensic analysis.</i>	U	C	Instructor-created exams / Seminar
CO3	<i>Understand the application of chemistry in agriculture and need of green methods</i>	U	C	Instructor-created exams / Assignment
CO4	<i>Understand the chemistry of soaps, synthetic detergents and their environmental effects.</i>	U	C	Instructor-created exams / Seminar
CO5	<i>Understand the chemistry of cosmetics and the effect on health.</i>	U	C	Instructor-created exams / Quiz

CO6	<i>Understand the chemistry of drugs, food additives their action and possible side effects</i>	U	C	Seminar/Viva
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Chemistry in Biological Systems & Forensic Chemistry		12	22
	1	Vitamins and Minerals: Name, source, function and deficiency diseases.	2	
	2	Enzymes - Classifications, characteristics, examples.	1	
	3	Hormones - Sex hormones - example, function. Pheromones.	2	
	4	Brain chemicals and human mood variations	1	
	5	General discussion of poisons with special reference to mode of action of cyanide, organophosphates and snake venom.	2	
	6	Detection of finger print, blood stain, semen, Breath analyzer	2	
	7	Sport doping-Steroids-Anabolic agents, Stimulants, Diuretics	2	
II	Chemistry and Agriculture		6	12
	8	Essential nutrients for plants – NPK value Chemical composition of soil, Soil enrichment	1	
	9	Fertilizers- natural, synthetic, mixed, NPK fertilizers. Excessive use of fertilizers and its impact on the environment. Bio fertilizers.	2	
	10	Pesticides: Classification – Insecticides, herbicides, rodenticides and fungicides (definition and examples only) – Non-degradable pesticides	2	
	11	Pesticide pollution and its impact on environment – Endosulfan disaster in Kerala (brief study).	1	
III	Cleansing agents and cosmetics		9	18
	12	Soaps – Hard and soft soaps – Alkali content – TFM – Detergents (classification) – Cleaning action – Advantages and disadvantages of soaps and detergents –	3	
	13	Shampoos: Ingredients and functions – Different kinds of shampoos (Antidandruff, anti-lice, herbal and baby shampoos).	1	
	14	Tooth paste: Composition and health effects. Hair dye: Chemicals used and its harmful effects.	1	
	15	Face and skin powders: Types, ingredients and functions. Cleansing creams: Cold creams, vanishing creams and bleach creams.	2	

	16	Perfumes, antiperspirants, sun screen preparations, nail polishes, lipsticks, rouges, eyebrow pencils and eye liners (ingredients and functions) – Harmful effects of cosmetics.	2	
IV	Pharmaceuticals and Dyes		9	18
	17	Drug: Chemical name, generic name and trade names with examples.	1	
	18	Terminology: Prodrug, pharmacy, pharmacology, pharmacophore, pharmacognosy, pharmacodynamics and pharmacokinetics (elementary idea only).	2	
	19	Antipyretics, analgesics, antacids, antihistamines, antibiotics, antiseptics, disinfectants, anaesthetics, tranquilizers, narcotics, antidepressants and psychedelic drugs (definition and examples).	2	
	20	Dyes: classification based on constitution, application, examples, uses.	2	
	21	Dyes: Requirements of a dye – Classification based on mode of application to the fabric –	1	
	22	Applications of dyes (general study). Ancient and modern colours – Mention of indigo and alizarin.	1	
V		Food Chemistry (OPEN ENDED)	9	
	23	Common adulterants Food Additives: Artificial sweeteners – Taste enhancers Artificial ripening of fruits and its side effects. Modern Food Habits:		

References

1. M. V. Kulkarni, *Biochemistry*, Pragati Books Pvt. Ltd., 2008.
2. S. C. Rastogi, *Biochemistry*, 2nd Edn., Tata McGraw Hill Publishing Co., New Delhi, 2007.
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4. N. V. Bhagavan, *Medical Biochemistry*, Academic Press, 2002.
5. Pharmaceutical Analysis, T. Higuchi and E.B. Hanseen, John Wiley and Sons, New York.
6. Quantitative Analysis of drugs, P.D. Sethi, 3rd edition, CBS Publishers, New Delhi, 1997.
7. Practical Clinical biochemistry methods and interpretations, R. Chawala, J.P. Brothers Medical Publishers (P) Ltd., 1995.
8. Laboratory manual in biochemistry, J. Jayaraman. New Age International Publishers, New Delhi, 1981.
9. H. S. Rathore, L. M. L. Nollet, *Pesticides: Evaluation of Environmental Pollution*, CRC Press, USA, 2012.
10. Murray Park, *The Fertilizer Industry*, Elsevier, 2001.
11. B. K. Sharma, *Industrial Chemistry*, Krishna Prakashan Media, 1991.
12. M. S. R. Winter, *A Consumer's Dictionary of Cosmetic Ingredients*, 7th Edn., Three Rivers Press, New York, 2009.
13. Gurdeep R. Chatwal, *Synthetic Drugs*, Himalaya Publishing House, Bombay, 1995.
14. Jayashree Ghosh, *A Textbook of Pharmaceutical Chemistry*, 3rd Edn., S. Chand and Company Ltd., New Delhi, 1999.
15. Lillian Hoagland Meyer, *Food Chemistry*, 1st Edn., CBS Publishers & Distributors, New Delhi, 2004.
16. B. A. Fox, A. G. Cameron, E. Arnold, *Food Science, Nutrition and Health*, 6th Edn., Edward Arnold, London, 1995.

17. A. Siddiqui, N. Anusha, *Deleterious Effects of Food Habits in Present Era*, J. Aller. Ther. 3:114, 2012.
18. H. S. Ramaswamy, M. Marcotte, *Food Processing: Principles and Applications*, CRC Press, 2005.
19. A. F. Smith, *Encyclopedia of Junk Food and Fast Food*, Greenwood Publishing Group, 2006.
20. T. A. M. Sagati, *The Chemistry of Food Additives and Preservatives*, John Wiley & Sons, 2012.
21. S. N. Mahindru, *Food Additives*, APH Publishing, 2009.
22. Biju Mathew, *Anchor India*, Info Kerala Communications Pvt. Ltd., 2015.

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	-	-	-	1	1	1			2	1		
CO 2	1		-	-	1	1	1			1	1		1
CO 3	-	-		1	2	2	1			2	2		1
CO 4	-	-			1	2	1			1	1	1	1
CO 5	-		-	1	2	2	1			2	2	1	1
CO 6	-	-	-	1	2	2	1			2	2	1	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/viva	Quiz/seminar/ Group discussion	End Semester Examinations
CO 1	✓		✓	✓
CO 2	✓		✓	✓
CO 3	✓	✓		✓
CO 4	✓		✓	✓
CO 5	✓		✓	✓
CO 6		✓	✓	

UNIVERSITY OF CALICUT
FIRST SEMESTER EXAMINATION
INORGANIC CHEMISTRY I
CHE1CJ101

Maximum Marks: 70

Duration : 2 hours

SECTION A (Short Answer)
Overall Ceiling 24
Answer all. Each question carry 3 marks

- | | |
|-----|---|
| 1. | Explain the involvement of chemistry in daily life with examples? |
| 2. | What do the terms absolute error and relative error mean with regard to analytical determinations? |
| 3. | Distinguish between the terms electronegativity and electron affinity? Explain their variation along a period and down a group? |
| 4. | Discuss the characteristics of ionic compounds and explain the factors affecting the formation of ionic bond? |
| 5. | AgCl is sparingly soluble in water while NaCl is soluble. Comment on this from lattice energy considerations? |
| 6. | Differentiate top-down and bottom-up approaches for the synthesis of nanomaterials? |
| 7. | Explain the significance of surface area to volume ratio in nanomaterial. Provide examples of how this ratio impacts the properties of nanomaterials? |
| 8. | Discuss the application of nanomaterials in electronics? |
| 9. | Discuss the importance of primary and secondary standards in volumetric analysis, providing examples of each? |
| 10. | Critically evaluate the advantages and limitations of the Double burette method of titration compared to other titration techniques? |

8x3= 24 Marks

SECTION B (Paragraph)
Overall Ceiling 36
Answer all. Each question carry 6 marks

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|-----|---|
| 11. | Explain the difference between accuracy and precision in analytical chemistry? Provide examples to illustrate each concept? |
| 12. | Describe the term standard deviation with respect to analytical determination? |

13.	Discuss the concept of isoelectronic species in the context of atomic and ionic radii. How does the nuclear charge affect the size of isoelectronic species?
14.	Discuss the conditions which favour covalent character in ionic compounds?
15.	Compare the bond length, bond energy and magnetic behavior of O_2 , O_2^+ , O_2^{2+} , O_2^- and O_2^{2-} with the help of Molecular Orbital Theory?
16.	Explain the classification of nanomaterials based on electron confinement?
17.	Explain the significance and applications of nanoparticles such as gold and silver nanoparticles in nanomaterials. How do size-dependent properties play a crucial role in their applications?
18.	Describe the safety measures and precautions that should be followed in a chemical laboratory. Discuss the importance of using Personal Protective Equipment (PPE) and handling hazardous chemicals safely?
6x6= 36 Marks	
SECTION C (Essay)	
Answer any one	
19.	Discuss Born-Haber cycle for NaCl? What are the applications of Born-Haber cycle?
20.	Briefly explain theory of adsorption and complexometric indicators?
1x10= 10 Marks	

UNIVERSITY OF CALICUT
FYUGP CHEMISTRY
FIRST SEMESTER EXAMINATION
CHE1MN101
BASIC INORGANIC AND NANOCHEMISTRY

Maximum Marks: 70

Duration : 2 hours

SECTION A (Short Answer)

Overall Ceiling 24

Answer all. Each question carry 3 marks

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|-----|--|
| 1. | Discuss the concept of orbit and orbital, highlighting the two major differences between them |
| 2. | Explain the significance of quantum numbers in atomic structure |
| 3. | Differentiate between ionic bond and covalent providing an example for each type |
| 4. | Discuss the Law of Triads and its significance in early attempts to classify elements |
| 5. | Elaborate on the Pauli's Exclusion Principle and how it influences the electron configuration of atoms |
| 6. | Explain the basic principles involved in complexometric titration. |
| 7. | Differentiate between Accuracy & Precision |
| 8. | Discuss Top-down processes and Bottom-up processes for the Synthesis of nanomaterials |
| 9. | Compare Fullerenes and graphene |
| 10. | Summarise the merits of Bohr Atom model |

8x3= 24 Marks

SECTION B (Paragraph)

Overall Ceiling 36

Answer all. Each question carry 6 marks

- | | |
|-----|---|
| 11. | Explain the shape of BeCl_2 , IF_7 , and XeF_2 using VSEPR theory |
| 12. | Explain bond order. How is it calculated? Give the significance of the bond order to explain the bond strength and bond length of a molecule |
| 13. | Illustrate with suitable equation – a) Molarity b) Mole fraction c) Normality |
| 14. | Explain solubility product and its applications in qualitative analysis |
| 15. | Discuss the significance of periodic properties in the modern periodic table and how Ionic radii, Electron affinity, and Oxidation number vary across periods and groups. |
| 16. | Demonstrate the classification of nanomaterials based on dimension with one example for each. |